

05-Angular-ModuleCoverage-Matrix: Campus Services Portal

This document serves as formal evidence of full Angular curriculum and framework capability coverage as specified in **BRD v2.0 (Angular) Section 8 (Frontend Requirements)** and **Section 12 (Deliverables)**.

Every concept listed below has an active, verified implementation file path within the `CampusServicesPortal1` project workspace.

1. Executive Summary & Build Verification

Metric	Status / Specification
Framework Version	Angular 19.2.1
Architecture	100% Standalone Components (<code>imports: [...]</code> , no legacy <code>NgModule</code>)
Build Status	<code>npx ng build</code> completes cleanly with 0 errors , production bundle verified
API Transport	Native <code>HttpClient</code> with typed Observables and JWT Interceptor
Component Total	104 Standalone Components across Admin, Student, Shared, and Layout
Service Total	27 Root-Injectable Typed Services
State Management	Angular Signals (<code>signal</code> , <code>computed</code> , <code>effect</code>) + RxJS

2. Core Angular Concepts Coverage Matrix

#	Concept / Feature Requirement	Implementation Evidence & File Path	Description in Campus Services Portal
1	Standalone Component Architecture	<code>Frontend/src/app/app.component.ts</code>	100% standalone structure without <code>NgModule</code> . Uses <code>imports: [CommonModule, RouterOutlet, NavbarComponent, ToastContainerComponent]</code> .
2	Application Bootstrap & Providers	<code>Frontend/src/app/app.config.ts</code>	<code>ApplicationConfig</code> bootstrapping <code>provideRouter(routes, withComponentInputBinding())</code> , <code>provideHttpClient(withInterceptors([authInterceptor, errorInterceptor]))</code> , and animations.
3	Route Parameters (<code>:id</code>)	<code>Frontend/src/app/app.routes.ts</code> <code>Frontend/src/app/features/student/lab-booking/student-booking.routes.ts</code>	Parameterized routes such as <code>/labs/:id/slots</code> , <code>/complaints/:id</code> , <code>/events/:id</code> , and <code>/fees/:id/receipt</code> . Extracted via <code>ActivatedRoute.params</code> and component input binding.
4	Query Parameters	<code>Frontend/src/app/features/admin/billing-management/pages/billing-dashboard/billing-dashboard.component.ts</code> <code>Frontend/src/app/features/admin/student-master/student-master.component.ts</code>	Linkable search, filter, and pagination states (e.g. <code>?search=&faculty=&page=1&pageSize=10</code>) synchronized with <code>router.navigate([], { queryParams: ... })</code> surviving page refreshes.
5	Child & Nested Routing	<code>Frontend/src/app/app.routes.ts</code> <code>Frontend/src/app/shared/layout-components/admin-layout/admin-layout.component.ts</code>	Dedicated shell layouts hosting child routes via nested <code><router-outlet></code> across Admin, Student, and Auth feature modules.

#	CONCEPT / FEATURE REQUIREMENT	IMPLEMENTATION EVIDENCE & FILE PATH	DESCRIPTION IN CAMPUS SERVICES PORTAL
6	Wildcard Redirection (404)	Frontend/src/app/app.routes.ts	Wildcard route { path: '**', component: NotFoundComponent } placed as the strict terminal entry in the route array.
7	Authentication Route Guard	Frontend/src/app/core/guards/auth.guard.ts	Functional canActivateFn verifying JWT presence, expiration, and redirecting unauthenticated users to /login with returnUrl.
8	Role-Based Authorization Guard	Frontend/src/app/core/guards/admin.guard.ts	Guard restricting access to /admin/** trees strictly to Admin and SuperAdmin roles, displaying a 403 Forbidden alert otherwise.
9	Lazy-Loaded Route Chunks	Frontend/src/app/app.routes.ts	Dynamic code splitting on every feature module using loadChildren and loadComponent: () => import(...).
10	Navigation Progress & Events	Frontend/src/app/shared/layout-components/navbar-component/navbar-component.component.ts	NavigationStart, NavigationEnd, NavigationCancel observables driving top progress bar and triggering unread notification refreshes.
11	Root Singleton Services	Frontend/src/app/core/services/auth.service.ts Frontend/src/app/core/services/notification.service.ts	Pure @Injectable({ providedIn: 'root' }) services injecting HttpClient, reading environment.apiUrl, and returning strongly-typed Observable<T>.
12	Reactive Forms Architecture	Frontend/src/app/features/auth/pages/register/register.component.ts Frontend/src/app/features/student/hostel-booking/pages/hostel-apply/hostel-apply.component.ts	Typed FormGroup, FormControl, and Validators managing comprehensive state validation before network submission.
13	Dynamic FormArray	Frontend/src/app/features/student/profile/components/edit-profile-modal/edit-profile-modal.component.ts Frontend/src/app/features/lab-shared/components/lab-grid-matrix/lab-grid-matrix.component.ts	Multi-row dynamic FormArray collections for telephone contact numbers, multiple addresses, and 2D workstation grid layouts.
14	Template-Driven Forms	Frontend/src/app/features/auth/pages/login/login.component.ts Frontend/src/app/features/admin/system-settings/pages/system-settings-page.component.ts	Pragmatic template-driven forms utilizing [(ngModel)], #form="ngForm", and inline validation directives.
15	Custom Field Validators	Frontend/src/app/core/validators/index-number.validator.ts Frontend/src/app/core/validators/campus-email.validator.ts	Custom regex validators verifying student index formats (PS/YYYY/NNNN) and university domain whitelists.
16	Cross-Field Custom Validation	Frontend/src/app/core/validators/password-match.validator.ts Frontend/src/app/features/student/hostel-booking/pages/hostel-apply/hostel-apply.component.ts	Multi-control validators ensuring password confirmation equality and hostel lease EndDate strictly following StartDate.
17	Custom Directives	Frontend/src/app/shared/directives/highlight.directive.ts Frontend/src/app/shared/directives/status-color.directive.ts	@Directive({ selector: '[appHighlight]' }) and [appStatusColor] manipulating element styling via HostListener and Renderer2.

#	CONCEPT / FEATURE REQUIREMENT	IMPLEMENTATION EVIDENCE & FILE PATH	DESCRIPTION IN CAMPUS SERVICES PORTAL
1 8	Custom Parameterized Pipes	Frontend/src/app/shared/pipes/short-text.pipe.ts Frontend/src/app/shared/pipes/status-label.pipe.ts Frontend/src/app/shared/pipes/time-range.pipe.ts	Custom pipes implementing <code>PipeTransform</code> : <code>shortText:maxLength</code> , <code>statusLabel</code> , and <code>timeRange</code> formatting time slots.
1 9	Built-in Angular Pipes	Frontend/src/app/features/admin/billing-management/pages/billing-dashboard/billing-dashboard.component.html	Extensive production usage of <code>currency: 'USD'</code> , <code>date: 'mediumDate'</code> , <code>uppercase</code> , <code>lowercase</code> , <code>number: '1.2-2'</code> , and <code>slice</code> .
2 0	Component Parent-Child Pattern	Frontend/src/app/features/admin/billing-management/components/assign-fee-modal/assign-fee-modal.component.ts	Strict parent-child separation: container components fetch and manage collections, passing immutable data downward via inputs and receiving action requests via <code>EventEmitter</code> outputs.
2 1	Four Screen State Pattern	Frontend/src/app/shared/components/loading-state Frontend/src/app/shared/components/error-state Frontend/src/app/shared/components/empty-state Frontend/src/app/shared/components/data-table	Every data-fetching view strictly handles 4 distinct UX states: Loading (skeleton/spinner), Error (retry card), Empty (call-to-action placeholder), and Data (active table/cards).
2 2	Destructive Action Confirmation	Frontend/src/app/shared/components/dialogs/confirm-modal/confirm-modal.component.ts	Reusable modal dialog intercepting deletions, fee assignments, workstation deactivations, and cancelations before API execution.
2 3	Reactive State Signals	Frontend/src/app/shared/components/status-badge/status-badge.component.ts Frontend/src/app/features/lab-shared/components/lab-grid-matrix/lab-grid-matrix.component.ts	Fine-grained reactivity using Angular Signals: <code>signal()</code> , <code>computed()</code> , and automatic effect triggers for counts, zoom scaling, and dynamic styling.
2 4	HTTP Interceptor Pipeline	Frontend/src/app/core/interceptors/auth.interceptor.ts Frontend/src/app/core/interceptors/error.interceptor.ts	Interceptors seamlessly injecting <code>Authorization: Bearer <token></code> into outgoing requests and catching 401/403/409 errors to trigger refresh tokens or toast alerts.
2 5	Responsive Design & Dark Theme	Frontend/tailwind.config.js Frontend/src/app/shared/layout-components/navbar-component/navbar-component.component.html	Complete Tailwind CSS responsive system supporting mobile drawers, desktop navbars, and automatic dark mode toggling.

3. Module Coverage Summary

BRD FUNCTIONAL MODULE	MODULE PATH	COMPONENTS	DEDICATED SERVICE	FORM IMPLEMENTATION
Module 1: Authentication & Identity	features/auth	4	AuthService	Reactive (Register, Reset) & Template (Login)
Module 2: Student Master & Profile	features/student/profile, features/admin/student-master	8	StudentProfileService, AdminStudentService	Reactive with FormArray (phones, addresses)

BRD FUNCTIONAL MODULE	MODULE PATH	COMPONENTS	DEDICATED SERVICE	FORM IMPLEMENTATION
Module 3: Hostel Management	features/student/hostel-booking , features/admin/hostel-management	10	HostelBookingService , AdminHostelService	Reactive with date-range custom validation
Module 4: Computer & Science Labs	features/student/lab-booking , features/admin/lab-management , features/lab-shared	14	LabBookingService , AdminLabService	Reactive 2D Workstation Grid Matrix & Time Slots
Module 5: Events & Venues	features/student/event , features/admin/event-management	8	StudentEventService , AdminEventService	Reactive Event Form & Capacity Slot Reservation
Module 6: Certificate Requests	features/student/certificate , features/admin/certificate-management	8	StudentCertificateService , AdminCertificateService	Reactive Form with document upload & copy counters
Module 7: Complaints & Feedback	features/student/complaint , features/admin/complaint-management	8	StudentComplaintService , AdminComplaintService	Dynamic Reactive Form keyed to category
Module 8: Notifications & Alerts	features/student/notification , features/admin/notification-monitor	6	NotificationService	Real-time Polling & Unread Status Badges
Module 9: Billing & Payment Ledger	features/student/billing , features/admin/billing-management	9	StudentPaymentService , AdminBillingService	Cohort Bulk Run & Single Fee Assignment
Module 10: Dashboards & Analytics	features/student/dashboard , features/admin/dashboard , features/admin/reports	11	StudentDashboardService , AdminDashboardService , ReportsService	Stat-Cards & Chart Visualization Panels
Module 11: Audit Logs & Admin Users	features/admin/audit-logs , features/admin/admin-management	8	AuditLogsService , AdminManagementService	Filter bar & Diff Inspection Modal
Module 12: System Administration	features/admin/system-settings , features/admin/faculty-management	6	SystemSettingsService , FacultyManagementService	Template & Reactive Global Config Forms